

Executive MBA Financial Accounting

Review Session 3

Angela Cheng

Credits to Collin Dursteler

CHICAGO BOOTH



The University of Chicago Booth School of Business

Housekeeping

Assignment 5 due Saturday!

- Start working earlier rather than later
- Deferred taxes can be difficult

Roadmap

- Leases today, deferred taxes Friday
- Available Thursday during day for help



Two views of leases...

No Capitalization:
Ongoing operating
cost

Lessor

Use of asset

Periodic transfers

Lessee

Rent payment

Capitalization:
In-substance
asset purchase

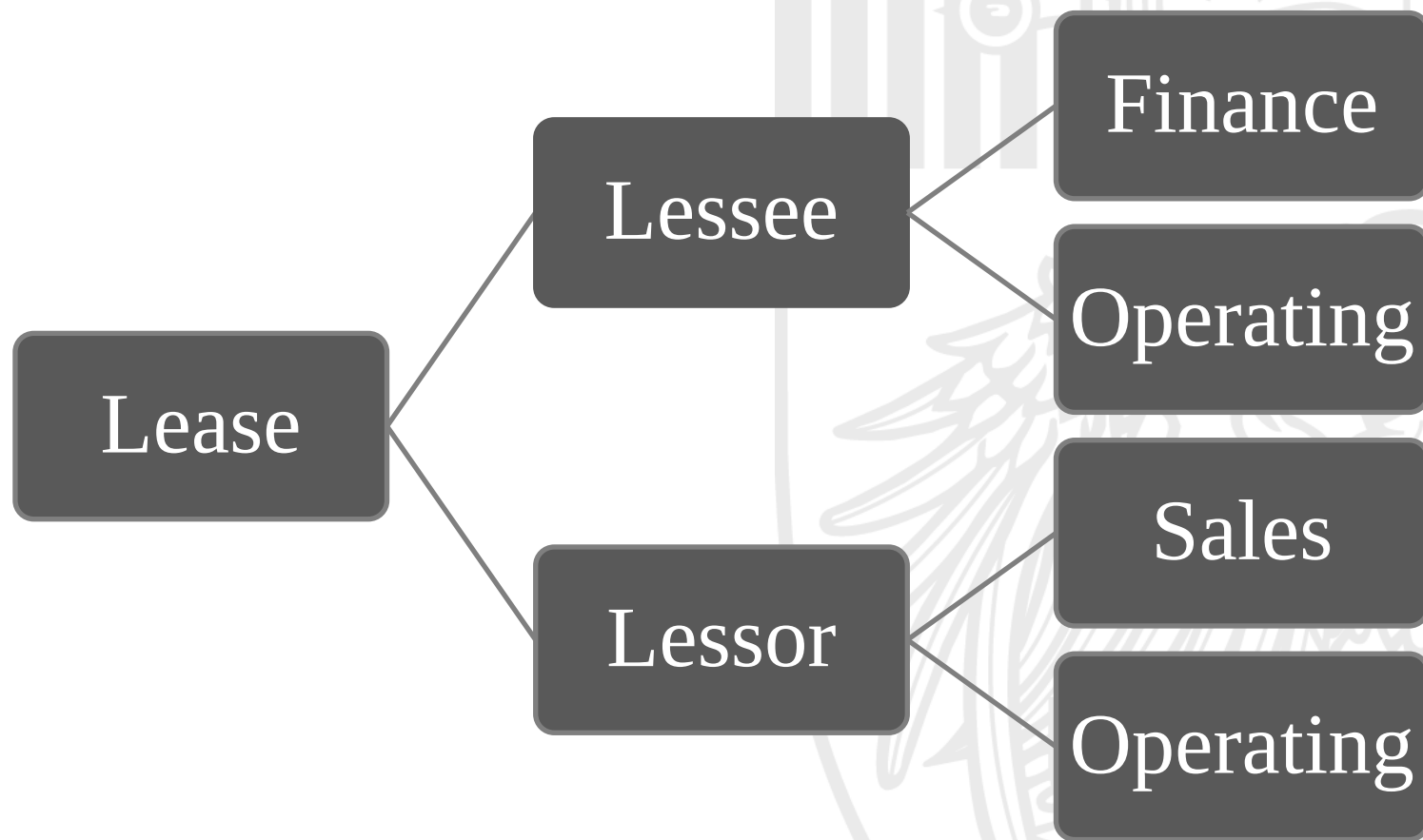
Lessor

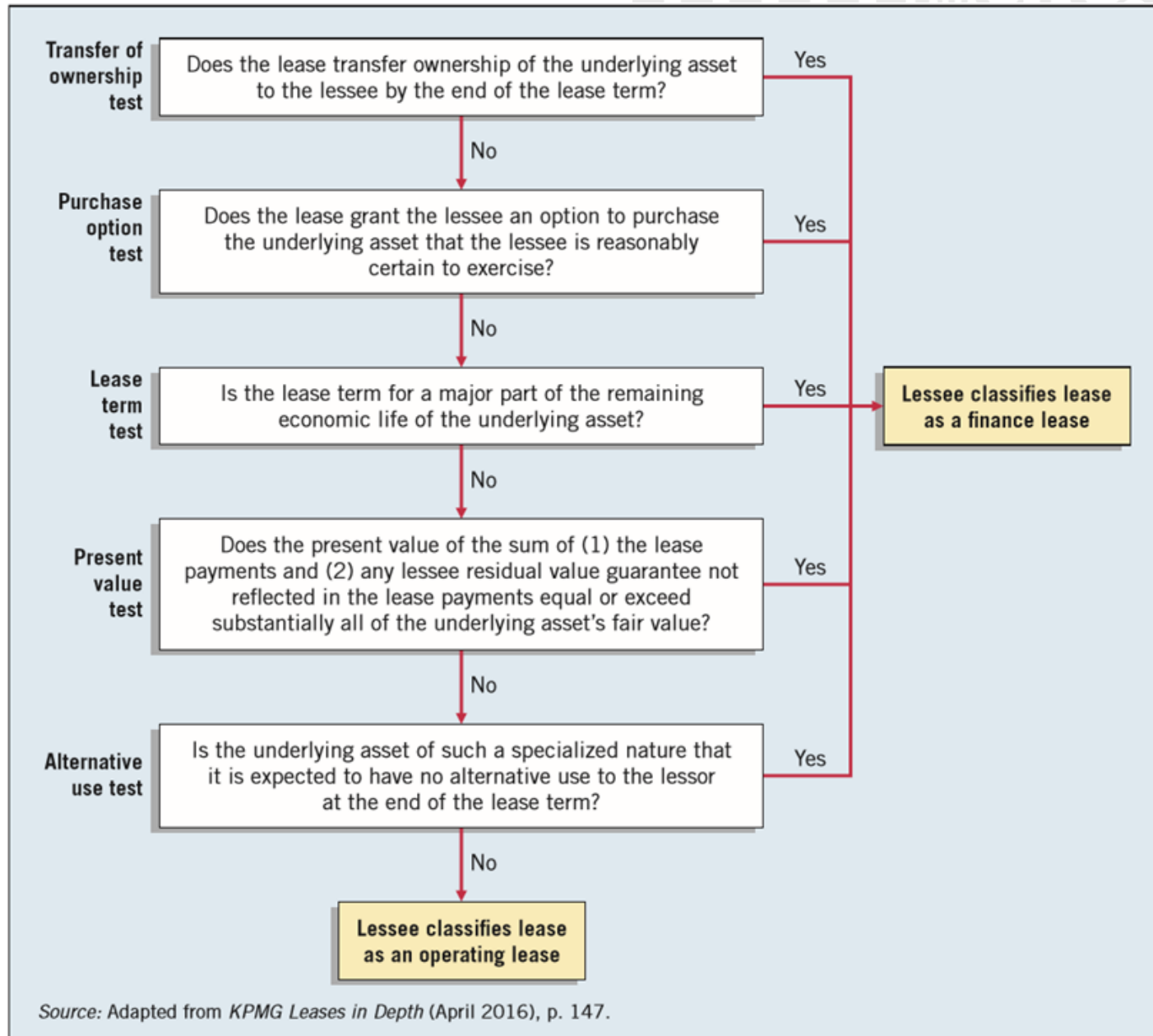
Asset

Transfer at lease inception

Lessee

Debt instrument





Three Entries to Remember

1. Start of lease contract

- Recognize transfer of control of asset (ROU Asset)
- Recognize liability to pay lessor (Lease Liability)
- Recorded at present value of all future lease payments
 - Use same formula for present value of an annuity

Dr. Right-of-Use Asset	\$XXX	
Cr. Lease Liability		\$XXX

Three Entries to Remember

2. Lease payments

- Represents contractual cash payments from lessee to lessor
- Cash payment is split into interest expense component and reduction of lease liability component
- Follows effective interest rate method

Dr. Interest Expense	\$XXX	
Dr. Lease Liability	\$XXX	
Cr. Cash		\$XXX

Three Entries to Remember

3. Asset amortization

- Like depreciation, reduces amount of capitalized asset recorded on Balance Sheet (ROU Asset)
- Recorded on a straight-line basis

Dr. Amortization Expense	\$XXX	
Cr. Right-of-Use Asset		\$XXX

Three Entries to Remember

1. Start of lease contract

- Recognize lease receivable
- Recognize revenue associated with transfer of asset

Dr. Lease Receivable	\$XXX	
Cr. Lease/Sales Revenue		\$XXX

Three Entries to Remember

2. Transfer of asset

- Because asset effectively transfers to lessee, remove asset from lessor books
- Recognize expense associated with *sale* of asset

Dr. Cost of Goods Sold	\$XXX	
Cr. Inventory/Asset		\$XXX

Three Entries to Remember

3. Rental payment receipt

- Like lessee treatment, record periodic receipt of contractual rent payment
- Split between interest and principal components

Dr. Cash	\$XXX	
Cr. Interest Revenue		\$XXX
Cr. Lease Receivable		\$XXX

Three Entries to Remember

1. Start of lease contract

- Recognize transfer of control of the use of asset (ROU Asset)
- Recognize liability to pay lessor (Lease Liability)
- Same treatment as financing lease at the inception

Dr. Right-of-Use Asset	\$XXX	
Cr. Lease Liability		\$XXX

Three Entries to Remember

2. Lease payments

- Represents contractual cash payments from lessee to lessor
- Now, entire cash payment amount is recorded as *Lease Expense* (no interest expense)

Dr. Lease Expense	\$XXX	
Cr. Cash		\$XXX

Three Entries to Remember

3. Asset and liability amortization

- Both right-of-use asset and lease liability are amortized at same rate
- Amortization amount comes from same effective interest rate table

Dr. Lease Liability	\$XXX	
Cr. Right-of-Use Asset		\$XXX

Two Entries to Remember

1. Lease payment receipt

- Cash payment received from lessee recorded as revenue

Dr. Cash	\$XXX	
Cr. Lease Revenue		\$XXX

Two Entries to Remember

2. Continue to depreciate asset

- Because lessor continues to maintain control of leased asset, continue to record amortization/depreciation

Dr. Depreciation Expense	\$XXX	
Cr. Acc. Depreciation		\$XXX

Practice 1, Lessee Corp.

CHICAGO BOOTH



The University of Chicago Booth School of Business

Lessee Corp. leased a satellite equipment from Lessor Corp. on 1/1/2019. The lease term is 3 years. The lease calls for payments of \$17,461.51 to be made at the end of each year. The incremental borrowing rate of the lessee is 8%. Assume that the lessor manufactured the satellite equipment at a cost of \$39,000.

- Let's first assume that **none of the finance lease criteria are met** and look at the operating lease treatment. Specifically, assume that the remaining economic life of the satellite is 10 years at 1/1/2019.
- Let's walk through all the relevant journal entries associated with the **operating lease**.

Lease Inception, Lessee

The lessee would recognize a right-of-use asset and lease liability at lease inception on 1/1/2019. The asset and liability are recorded at the **present value** of all future lease payments; i.e., \$45,000.

$$PVANN_{3,8\%} = \frac{[1 - (1 + 0.08)^{-3}]}{0.08} = 2.577$$

So, present value of three payments of \$17,461.51 is
 $\$17,461.51 * 2.577 = \$45,000$.

Lease Inception, Lessee

The lessee would recognize a right-of-use asset and lease liability at lease inception on 1/1/2019. The asset and liability are recorded at the **present value** of all future lease payments; i.e., \$45,000.

Dr. Right-of-Use (ROU) Asset	\$45,000	
Cr. Lease Liability		\$45,000

Lease Payments, Lessee

At the end of each year, the lessee recognizes lease expense for the lease payment.

- This is calculated using the *effective interest method*.
- Do not be confused by this table: While it is more useful for finance leases, we obtain the **amount of ROU asset/lease liability amortization** from the schedule.
- Be comfortable with how to construct these schedules.

Lease Payments, Lessee cont'd

Year (1)	Beg. Balance (2)	Interest Exp. (3) = (2)*8%	Cash Payment (4)	Reduction in Principal (5) = (4) – (3)	Ending Balance (6) = (2) – (5)
2019	\$45,000.00	\$3,600.00	\$17,461.51	\$13,861.51	\$31,138.49
2020	31,138.49	2,491.08	17,461.51	14,970.43	16,168.06
2021	16,168.06	<u>1,293.45</u>	<u>17,461.51</u>	<u>16,168.06</u>	0
	Total	<u>\$7,384.53</u>	<u>\$52,384.53</u>	<u>\$45,000.00</u>	

Years 1–3		
Dr. Lease Expense	\$17,461.51	
Cr. Cash		17,461.51

Lease Amortization, Lessee

Year 1		
Dr. Lease Liability	\$13,861.51	
Cr. Right-of-Use Asset		13,861.51

Year 2		
Dr. Lease Liability	\$14,970.43	
Cr. Right-of-Use Asset		14,970.43

Year 3		
Dr. Lease Liability	\$16,168.06	
Cr. Right-of-Use Asset		16,168.06

Lessor Entries

The lessor *does not* derecognize the asset and so no gain/loss is recognized at the inception of the lease. Instead, the lessor treats the contract as a long-term rental contract starting 1/1/2019.

Lessor Entries, cont'd

On Dec. 31st of each of the next three years, the *lessor* would make the following entry to recognize the receipt cash lease payments.

Dr. Cash	\$17,461.51	
Cr. Lease Revenue		17,461.51

Lessor Entries, cont'd

The lessor continues to recognize depreciation expense on the leased equipment. $\$39,000/10 = \$3,900$ each year-end.

Dr. Depreciation Expense	\$3,900	
Cr. Acc. Depreciation		3,900

Lease Inception, Lessee

The lessee would recognize a right-of-use asset and lease liability at lease inception on 1/1/2019. The asset and liability are recorded at the **present value** of all future lease payments; i.e., \$45,000.

Dr. Right-of-Use (ROU) Asset	\$45,000	
Cr. Lease Liability		\$45,000

ROU Asset Amortization, Lessee

Right-of-use asset is amortized over the term of the lease contract with a straight-line method ($45,000/3 = 15,000/\text{year}$).

- Now that the right to control the asset has effectively transferred from the lessor to the lessee, the lessee records the capitalized asset's depreciation/amortization.

Years 1–3		
Dr. Amortization Exp.	\$15,000	
Cr. Right-of-Use Asset		15,000

Lease Payments, Lessee

At the end of each year, the lessee recognizes interest expense and the reduction in lease liability using the *effective interest method*.

- Same table seen previously in operating lease calculations
- Be comfortable with how to construct these schedules.

Year (1)	Beg. Balance (2)	Interest Exp. (3) = (2)*8%	Cash Payment (4)	Reduction in Principal (5) = (4) – (3)	Ending Balance (6) = (2) – (5)
2019	\$45,000.00	\$3,600.00	\$17,461.51	\$13,861.51	\$31,138.49
2020	31,138.49	2,491.08	17,461.51	14,970.43	16,168.06
2021	16,168.06	<u>1,293.45</u>	<u>17,461.51</u>	<u>16,168.06</u>	0
	Total	<u>\$7,384.53</u>	<u>\$52,384.53</u>	<u>\$45,000.00</u>	

Lease Payments, Lessee cont'd

Year 1		
Dr. Interest Expense	\$3,600	
Dr. Lease Liability	13,861.51	
Cr. Cash		17,461.51

Year 2		
Dr. Interest Expense	\$2,491.08	
Dr. Lease Liability	14,970.43	
Cr. Cash		17,461.51

Lease Payments, Lessee cont'd

Year 3		
Dr. Interest Expense	\$1,293.45	
Dr. Lease Liability	16,168.06	
Cr. Cash		17,461.51

Lease Inception, Lessor

The lessor would recognize a lease receivable and lease revenue lease inception on 1/1/2019. Additionally, the lessor would remove the leased asset from its book as control of the asset has passed to the lessee. The lease receivable is recorded at the **present value** of all future lease payments; i.e., \$45,000.

Dr. Lease Receivable	\$45,000	
Cr. Lease Revenue		\$45,000
Dr. Cost of Goods Sold	39,000	
Cr. Leased Inventory/Equip.		39,000

Lease Payments, Lessor

At the end of each year, the lessee recognizes interest expense and the reduction in lease liability using the *effective interest method*.

- Same table seen previously in operating lease calculations
- Be comfortable with how to construct these schedules.

Year 1		
Dr. Cash	\$17,461.51	
Cr. Interest Revenue		3,600
Cr. Lease Receivable		13,861.51

Lease Payments, Lessor cont'd

Year 2		
Dr. Cash	\$17,461.51	
Cr. Interest Revenue		2,491.08
Cr. Lease Receivable		14,970.43

Year 3		
Dr. Cash	\$17,461.51	
Cr. Interest Revenue		1,293.45
Cr. Lease Receivable		16,168.06

Practice 2, Southwest Airlines

CHICAGO BOOTH



The University of Chicago Booth School of Business

The following information is extracted from Southwest Airlines fiscal 2019 annual report regarding its leases:

As of December 31, 2019, maturities of lease liabilities were as follows:

<u>(in millions)</u>	<u>Operating leases</u>	<u>Finance leases</u>
2020	\$ 392	\$ 107
2021	257	102
2022	141	98
2023	113	94
2024	92	90
Thereafter	661	230
Total lease payments	\$ 1,656	\$ 721
Less imputed interest	(325)	(94)
Total lease obligations	1,331	627
Less current obligations	(353)	(85)
Long-term lease obligations	<u>\$ 978</u>	<u>\$ 542</u>

CHICAGO BOOTH Practice Problem, Southwest cont'd

- At what amount is the **total lease obligation** for operating and finance leases recognized on Southwest's balance sheet at December 31, 2019?

❖ $1,331 + 627 = \mathbf{1,958}$

As of December 31, 2019, maturities of lease liabilities were as follows:

<u>(in millions)</u>	<u>Operating leases</u>	<u>Finance leases</u>
2020	\$ 392	\$ 107
2021	257	102
2022	141	98
2023	113	94
2024	92	90
Thereafter	661	230
Total lease payments	\$ 1,656	\$ 721
Less imputed interest	(325)	(94)
Total lease obligations	1,331	627
Less current obligations	(353)	(85)
Long-term lease obligations	\$ 978	\$ 542

CHICAGO BOOTH Practice Problem, Southwest cont'd

2. Assuming the lease payments for finance leases after 2024 are made evenly over 2 years. What is the amount of long-term lease obligations with respect to outstanding Finance leases at December 31, 2019 that Southwest recognized on its balance sheet at the end of fiscal 2020, i.e., December 31, 2020?

Step (a): Find out the current and non-current portion of the lease obligations recognized at December 31, 2019 from the information provided.

- At December 31, 2019, the current finance lease obligation would be \$85 (paid in fiscal 2020), and the non-current finance lease obligation would be $\$627 - \$85 = \$542$.
- This \$542 is also the total outstanding lease obligation at the end of fiscal 2020, i.e., December 31, 2020.

Step (b): Find out the incremental borrowing rate to calculate the interest expense during fiscal 2021.

	12/31/2019	12/31/2020	12/31/2021	12/31/2022	12/31/2023	12/31/2024	12/31/2025	12/31/2026
	-627	107	102	98	94	90	115	115
IRR	3.57%							

From (a), we know the total outstanding lease obligations at the beginning of 2021 i.e., at January 1, 2021 is \$542. To arrive at the interest expense of lease liability during fiscal 2021:

$$\begin{aligned}\text{Interest expense} &= \text{carrying value of the lease liability} \times \text{interest rate} \\ &= \$542 \times 3.57\% = \$19\end{aligned}$$

Dr. Interest Expense (calculated)	\$19	
Dr. Lease Liability (plug)	\$83	
Cr. Cash (given)		\$102

Step (c): From (b), we know the current portion of the outstanding finance lease obligations at 12/31/2020 is \$83 (\$83 will be paid down in fiscal 2021). Therefore, the non-current portion at 12/31/2020 is total outstanding finance lease obligations minus current-portion ($\$542 - \$83 = \$459$).

CHICAGO BOOTH Practice Problem, Southwest cont'd

3. (a) What is the expected impact, if any, of the lease payments of \$107 million and \$392 million made by Southwest with respect to **financing and operating Leases**, respectively, on Southwest's earnings before taxes (EBT) during fiscal 2020?

As of December 31, 2019, maturities of lease liabilities were as follows:

<u>(in millions)</u>	Operating leases	Finance leases
2020	\$ 392	\$ 107
2021	257	102
2022	141	98
2023	113	94
2024	92	90
Thereafter	661	230
Total lease payments	\$ 1,656	\$ 721
Less imputed interest	(325)	(94)
Total lease obligations	1,331	627
Less current obligations	(353)	(85)
Long-term lease obligations	\$ 978	\$ 542

3. (b) What is the expected impact, if any, of the lease payments of \$107 million and \$392 million made by Southwest with respect to **financing and operating Leases**, respectively, on Southwest's earnings before taxes (EBT) during fiscal 2020?

FINANCING LEASES		
Dr. Interest Expense	\$22	
Dr. Lease Liability	85	
Cr. Cash		\$107
OPERATING LEASES		
Dr. Lease Expense	\$392	
Cr. Lease Cash		392

- ❖ Lease payments **decrease EBT by $22 + 392 = \$414$ million.**
- ❖ While the ROU Asset amortization entry *also* affects EBT (Dr. Amortization Exp. and Cr. ROU), this is a non-cash expense.

CHICAGO BOOTH Practice Problem, Southwest cont'd

4. (a) What is the impact of the cash lease payment of \$392 million made by Southwest with respect to **operating leases** on its statement of cash flows (CFI, CFF, and CFO) for fiscal 2020?
 - ❖ For operating leases, the cash payment is an operating cash flow: **CFO decreases by \$392 million, CFI and CFF are unaffected.**

As of December 31, 2019, maturities of lease liabilities were as follows:

<u>(in millions)</u>	<u>Operating leases</u>	<u>Finance leases</u>
2020	\$ 392	\$ 107
2021	257	102
2022	141	98
2023	113	94
2024	92	90
Thereafter	661	230
Total lease payments	\$ 1,656	\$ 721
Less imputed interest	(325)	(94)
Total lease obligations	1,331	627
Less current obligations	(353)	(85)
Long-term lease obligations	\$ 978	\$ 542

CHICAGO BOOTH Practice Problem, Southwest cont'd

4. (b) What is the impact of the cash lease payment of \$107 million made by Southwest with respect to **financing leases** on the statement of cash flows (CFI, CFF, and CFO) for fiscal 2020?

As of December 31, 2019, maturities of lease liabilities were as follows:

<u>(in millions)</u>	<u>Operating leases</u>	<u>Finance leases</u>
2020	\$ 392	\$ 107
2021	257	102
2022	141	98
2023	113	94
2024	92	90
Thereafter	661	230
Total lease payments	\$ 1,656	\$ 721
Less imputed interest	(325)	(94)
Total lease obligations	1,331	627
Less current obligations	(353)	(85)
Long-term lease obligations	\$ 978	\$ 542

CHICAGO BOOTH Practice Problem, Southwest cont'd

4. (b) What is the impact of the cash lease payment of \$107 million made by Southwest with respect to **financing leases** on the statement of cash flows (CFI, CFF, and CFO) for fiscal 2020?

Dr. Interest Expense (plug)	\$22	
Dr. Lease Liability (given)	\$85	
Cr. Cash (given)		\$107

- ❖ The principal component represents a financing cash flow: **CFF decreases by \$85 million**. The interest component represents an operating cash flow: **CFO decreases by \$22 million. CFI is unaffected.**